

Steven Gough-Kelly

PHD · MSc (BY RESEARCH) · BSc (HONS) · MINSTP · FRAS ·

✉ sgoughkelly@gmail.com | 🏠 steventgk.github.io/ | 📧 steventgk | 🌐 steventgk | 🐦 SGKastro

My primary research is studying the formation of Box/Peanut bulges in barred galaxies by comparing isolated and cosmological simulations to observations of external galaxies and the Milky Way. Recent work has focused on exploring the structure and kinematics of differently aged populations in the bar/bulge of such galaxies as indicators for evolutionary histories.

Education

Jeremiah Horrocks Institute, University of Central Lancashire

Preston, UK

PHD GALAXY DYNAMICS

2020 - 2024

- Thesis Title: Kinematic Fractionation of the Stellar Populations in Barred Galaxies
- Advisor: Prof. Victor P. Debattista
- Awarded *Moses Holden Studentship*

Jeremiah Horrocks Institute, University of Central Lancashire

Preston, UK

MSc (BY RESEARCH) ASTROPHYSICS

2018 - 2019

- Thesis Title: Proper Motions of Bulge Stars
- Advisor: Prof. Victor P. Debattista

University of Central Lancashire

Preston, UK

BSc (HONS) ASTROPHYSICS

2015 - 2018

- Dissertation Title: Evolution of Orbits That Support a Nuclear Disc
- Advisor: Prof. Victor P. Debattista
- Awarded School Prize for Best Third Year Astrophysics Project

Experience

Westholme School

Blackburn, UK

SENIOR SCIENCE TECHNICIAN

Jan 2024 - March 2025

- Plan, prepare and facilitate science experiments across the school for all three sciences.
- Audit science equipment and facilities.
- Successful implementation of new request and inventory software.
- Implement safe working practices.
- Training of Junior Technicians.
- Writing and updating department policies.

Westholme School

Blackburn, UK

TEACHER OF PHYSICS

November 2023

- Teaching of physics curriculum across classes aged 11-18 years old including exam classes.

University of Central Lancashire

Preston, UK

MARKET INSIGHTS DATA ANALYST

Jul 2023 - Feb 2024

- Develop custom pipeline and workflow for data.
- Deploy custom dashboards for inquiry.
- Consult with the wider team on improvements and new metrics.

Freelance Production Engineer

UK

LIGHTING, AUDIO, STAGE, VIDEO, SFX, MANAGEMENT, SITE

2016 - present

- Various roles and responsibilities across a wide variety of live entertainment productions.

Publications

ORCID: 0000-0003-4799-5079

PUBLISHED

- San Martin Fernandez L. M., **Gough-Kelly S.**, Debattista V. P. Gonzalez O. A., Lazar I., Rojas-Arriagada A., Beraldo e Silva L. (2025) *The chemical and spatial variations of the bulge's velocity ellipsoids*, *MNRAS*, doi:10.1093/mnras/staf887
- Prudil Z., Debattista V. P., Beraldo e Silva L., Anderson S. R., **Gough-Kelly S.**, Kunder A., Rejkuba M., Gerhard O., Wyse R. F. G., Koch-Hansen A. J., Rich D. M., Savino A. (2025) *The Galactic Bulge exploration V.: The secular spherical and X-shaped Milky Way bulge*, *A&A*, doi:10.1051/0004-6361/202554819
- Prudil Z., Kunder A., Beraldo e Silva L., **Gough-Kelly S.**, Rejkuba M., Anderson S. R., Debattista V. P., Gerhard O., Rich D. M., Nataf A. J., Koch-Hansen A. J., Savino A., Dekany I., (2025) *The Galactic Bulge exploration IV.: RR Lyrae stars as traces of the Galactic bar – 3D and 5D analysis, extinction variation*, *A&A*, doi:10.1051/0004-6361/202450620
- Anderson S. R., **Gough-Kelly S.**, Debattista V. P., Du M., Erwin P., Cuomo V., Caruana J., Hernquist L., Vogelsberger M. (2024), *The interplay between accretion, galaxy downsizing and the formation of box/peanut bulges in TNG50*, *MNRAS*, doi:10.1093/mnras/stad3271
- Gough-Kelly S.**, Debattista V. P., Clarkson W. I., Gonzalez O. A., Anderson S. R., Gennaro M., Calamida A., Sahu, K. C., 2022, *Predicted trends in Milky Way bulge proper motion rotation curves: future prospects for HST and LSST*, *MNRAS*, 509, 4, 4829–4848. doi:10.1093/mnras/stab3192

SUBMITTED

- Fiteni K., Anderson S. R., Debattista V. P., Caruana J., Amarante J. A. S., **Gough-Kelly S.**, Eyer L., Beraldo e Silva L., Khachataryants T., Cuomo V. (2025) *The Edge of the Milky Way's Star-Forming Disc: Evidence from a "U-Shaped" Stellar Age Profile*, submitted to *A&A*

IN PREP

- Gough-Kelly S.**, Debattista V. P., Eyer L., Anderson S. R., Fiteni K., Amarante J. A. S., Khachataryants T., Caruana J., Beraldo e Silva L., Cuomo V. (2025) *Characterising Miras in Gaia: Evidence of a Peanut-Shaped Age Dispersion*
- Gough-Kelly S.**, Anderson S. R., Debattista V. P., Du M. (2026) *Kinematic Moments of the Stellar Populations in Barred Galaxies*
- M^cDonald B. F., Debattista V. P., **Gough-Kelly S.**, Prudil Z., San Martin Fernandez L. M. (2026) *Kinematic Fractionation in the Galactic Bulge: Evidence from Proper Motions*

WHITE PAPERS

- Kruszyńska, K., Street, R. A., **Gough-Kelly, S.**, Bonito, R., Prisinzano, L., Trivedi, O., Gandhi, P., Hundertmark, M., Tsapras, Y., Di Criscienzo, M., Musella, I., Dall'Ora, M., Bachelet, E., Abrams, N. S., Khakpash, S., Rabus, M., Szkody, P., Holt, C., 2024, *Synergies between Roman Galactic Plane Survey and other major surveys*, *arXiv*, astro-ph.GA, arXiv:2406.14767
- Street, R. A., **Gough-Kelly, S.**, Lam, C., Varela, A., Makler, M., Bachelet, E., Lu, J. R., Abrams, N., Pusack, A., Terry, S., Di Stefano, R., Tsapras, Y., Hundertmark, M. P. G., Grand, R. J. J., Daylan, T., Sobeck, J., 2023, *Maximizing science return by coordinating the survey strategies of Roman with Rubin, and other major facilities*, *arXiv*, astro-ph.IM, doi:10.48550/arXiv.2306.13792

VARIA

- Gough-Kelly S.**, Street, R. A., (2022), *Roman with Rubin*, *Astronomy & Geophysics (RAS)*, 64, 5.12. doi:10.1093/astrogeo/atad043
- Maunder M., O'Brien Á., Reid J., Bowman D. M., Richards F., **Gough-Kelly S.**, (2022), *Generation Covid*, *Astronomy & Geophysics (RAS)*, 63, 3.22. doi:10.1093/astrogeo/atac037

Presentations

INVITED TALKS

- 2025 *Heteroscedastic Observations of Large-Amplitude Long-Period Variables*, Space Astronomy Science Platforms Focus Week.
- 2025 *Galaxies with the Next Generation of Simulations and Great Observatories*, Mexborough & Swinton Astronomical Society, UK.
- 2023 *Simulating the Internal Evolution of Barred Galaxies*, Loebman Lab UC Merced, CA, USA.
- 2023 *The Next Generation of Barred Galaxies*, TAPIR Caltech, CA, USA.
- 2022 *The Next Generation of Barred Galaxies*, University of Nottingham, UK.
- 2022 *Galaxy Evolution in the Era of Gaia and IllustrisTNG*, Preston and District Astronomical Society, UK.
- 2021 *What the $H\epsilon \subset \kappa$? : A Dyslexic Researcher's Review of Accessibility*, Association of Learned and Professional Society Publishers - University Press Redux, UK.
- 2020 *Modelling the Milky Way in the Era of Gaia*, California State University Fresno, CA, USA.

CONTRIBUTED TALKS

- 2023 *Modelling the Internal Evolution of Barred Galaxies*, AAS DDA 54th Annual Meeting, Virtual
- 2023 *Modelling the Internal Evolution of Barred Galaxies*, UCLan Postgraduate Research Conference, UK.
- 2022 *Predicted trends in Milky Way bulge proper motion rotation curves: future prospects for HST and LSST*, LSST@Europe4, Italy.
- 2022 *Cosmic Peanuts - Kinematic Fractionation in TNG50 B/P Galaxies*, N-Body Workshop, Flatiron Institute, NY, USA.
- 2022 *Cosmic Peanuts - Galaxy Dynamics in the Era of IllustrisTNG*, Jeremiah Horrocks Institute CARD Talks, UK.
- 2021 *Age Dissection of Milky Way Bulge Kinematics*, N-Body Shop Excellence Conference, Virtual
- 2021 *Cosmic Peanuts - Kinematic Fractionation in the Era of IllustrisTNG*, Jeremiah Horrocks Institute CARD Talks, UK.
- 2020 *Separation of Bulge Kinematics*, Jeremiah Horrocks Institute CARD Talks, UK.
- 2019 *Proper Motions of Bulge Stars*, Jeremiah Horrocks Institute CARD Talks, UK.

PUBLIC TALKS

- 2024 *Colours and Sounds of Space*, Westholme School, UK.
- 2023 *2-Space 2-Gay*, UCLan Students' Union, UK.
- 2022 *Space is Gay*, UCLan Students' Union, UK.
- 2021 *What the $H\epsilon \subset \kappa$? : A Dyslexic Researcher's Review of Accessibility*, Cambridge University Press, UK.

POSTERS

- 2024 *Biases in Weighted Estimation Methods* (L. Eyer, S. Gough-Kelly, D. Evans, L. Rimoldini), EAS2024, Padova, Italy.
- 2023 *Predicted trends in Milky Way bulge proper motion rotation curves: future prospects for HST and LSST* (S. Gough-Kelly), AAS241 Seattle, WA, USA.
- 2017 *Developing Astronomy Planetarium Shows with Stellarium* (S. Gough-Kelly, D. Brown), UCLan Research Internship Program Presentation, UK.

Conferences & Workshops Organisation

Space Astronomy Science Platforms Focus Week (Workshop)

ESA GAIA & EUCLID

- **Invited Speaker:** *Heteroscedastic Observations of Large-Amplitude Long-Period Variables*

Virtual

2025

Rubin Community Workshop

RUBIN OBSERVATORY

Delegate

Virtual

2024

DDA54

AAS DIVISION FOR DYNAMICAL ASTRONOMY

- **Slack Chair:** Special Session: Accessibility and Inclusivity in the Dynamics Community
- **Oral Presentation:** *Modelling the Internal Evolution of Barred Galaxies*

Virtual

2023

AAS241

AMERICAN ASTRONOMICAL SOCIETY

- **Session Chair:** Simulating Stars and Gas in Galaxies
- **iPoster Presentation:** *Predicted trends in Milky Way bulge proper motion rotation curves: future prospects for HST and LSST*

Seattle Convention Center, USA

2023

National Astronomy Meeting

ROYAL ASTRONOMICAL SOCIETY

- **Organizer:** Early Career Researchers Parallel Session
- **Chair & Organizer:** Early Careers Networking Town Hall
- **Poster Presentation:** *Predicted trends in Milky Way bulge proper motion rotation curves: future prospects for HST and LSST*

University of Warwick, UK

2022

Understanding Galaxy Evolution Using Simulations (Workshop)

GALACTIC FIDELITY SEMINAR SERIES

- **Presenter:** Exercises on Understanding Galaxy Evolution Using Simulations
- Created and presented solutions to workshop activities on manipulating and analyzing galaxy simulations.

Virtual

2021

Introduction to $\text{\LaTeX}$ (Workshop)

UCLAN PHYSICS SOCIETY

- **Presenter:** Development and presentation of an introductory course on $\text{\LaTeX}$ for scientific report writing and publication.

UCLan, UK

2018

Teaching Experience

UNDERGRADUATE COURSES

Over five years of experience as a Teaching Assistant and marker for undergraduate courses in physics, astronomy, scientific computing and practical laboratory classes, including remote classes. Tasks have included setting up equipment, grading, formative feedback on assignments, mentoring projects, radiation supervision and improving/developing experiments. Group teaching of practical skills, presentation giving and report writing.

2022-2023	Foundations of Applied Physics , Foundation Degree	UCLan, UK
fall 2022	Practical Physics 1 - Programming and Experimental Physics , Year 1 Undergraduate	
2021-2022	The Milky Way , Year 2 Undergraduate	
2020-2021	Practical Physics2 - Computing and Experimental Physics , Year 2 Undergraduate	
2020-2021	Cosmology and Galaxies , Year 3 Undergraduate	
2020-2021	Galaxies Beyond the Milky Way , Year 2 Undergraduate	
2019-2020	The Milky Way , Year 2 Undergraduate	
2018-2020	Year 2 Laboratory , Year 2 Undergraduate	

HIGH-SCHOOL

Teacher of Physics

WESTHOLME SCHOOL, UK

Blackburn, UK

winter 2023

Taught physics for four weeks to students aged 11 to 18, including exam classes.

Teach Physics Internship

LANCASTER GIRLS GRAMMAR SCHOOL, UK

Ogden Trust, UK

summer 2017

Selected for a competitive internship through the Ogden Trust as a funded teaching assistant at Lancaster Girls Grammar School (UK) to inspire young students to study physics. Tasks included marking, lesson planning, delivery of national curriculum lessons, and targeted physics career workshops.

Mentorship

Ben M.

PHD

ULan, UK

2025

Writing Journal articles. Studying the Milky Way using Simulations

Stuart A.

PHD

ULan, UK

2023-2025

Writing Journal articles. Studying the Milky Way using Simulations. Advanced Python.

Ben M.

UNDERGRADUATE

UCLan, UK

2023-2024

Introduction to galaxy dynamics using simulations with Python programming. Analysis of the TNG50 simulation using large-scale computing.

Maria A.

UNDERGRADUATE

University of Malta

2022-2023

Introduction to galaxy dynamics using simulations with Python programming and data visualizations for a dissertation project.

Luis S. M. F.

POSTGRADUATE (MSc BY RESEARCH)

UCLan, UK

2021-2022

Introduction to galaxy dynamics and the Milky Way using simulations for a dissertation project.

William R.

UNDERGRADUATE

UCLan, UK

2019-2020

Demonstrating the Moses Holden Telescope (UCLan) and image data reduction. Supervised and assisted multi-band imaging of asteroids. Their dissertation project won the school prize for the best research project.

Awards, Fellowships, & Grants

2023	Ogden Outreach Award , Ogden Trust, UK	£ 25 / \$ 30
2023	Oral Presentation 3rd Prize , UCLan Postgraduate Research Conference	£ 300 / \$ 390
2023	Travel Funds , Jeremiah Horrocks Institute, University of Central Lancashire	£ 3000 / \$ 3900
2022	RAS Grants , Royal Astronomical Society, UK	£ 500 / \$ 650
2022	Conference Fund , Jeremiah Horrocks Institute, University of Central Lancashire	£ 500 / \$ 650
2020	Moses Holden Studentship (Ph.D.) , Jeremiah Horrocks Institute, University of Central Lancashire	
2019	Alumni Fund , Ogden Trust, UK	£ 900 / \$ 1200
2018	School Prize for Best Third Year Astrophysics Project , University of Central Lancashire, UK	
2017	UCLan Through the Ages Grant , University of Central Lancashire Students' Union	£ 200 / \$ 260
2017	Research Internship , Ogden Trust, UK	£ 1800 / \$ 2300
2017	Teach Physics Internship , Ogden Trust, UK	£ 1500 / \$ 2000
2016	Science Demonstration Development Fund , University of Central Lancashire	£ 200 / \$ 260
2016	Research Internship , Ogden Trust, UK	£ 1500 / \$ 2000
2014	School Physicist of the Year (Oldham Sixth Form College) , Ogden Trust, UK	

Observing Experience

Moses Holden Telescope

UCLan, UK

20 NIGHTS OF OBSERVING EXPERIENCE

2018-2020

Observing deep-sky and variable targets for undergraduate laboratory experiments and demonstrating the 0.7 m Planewave CDK700 Telescope to students.

Moses Holden Telescope

UCLan, UK

5 NIGHTS OF OBSERVING SUPERVISION

winter 2019

Supervising and supporting the multi-band imaging the passage of a nearby asteroid 37 Fides, to determine the spectral energy distribution for an undergraduate dissertation.

Moses Holden Telescope

UCLan, UK

5 NIGHTS COMMISSIONING OBSERVATIONS

winter 2016

Support in building the telescope pointing model by manually aligning targets with the World Coordinate System of the telescope.

Training

Tableau Essential Training

Online

LINKEDIN LEARNING

2023

Certified by the National Association of State Boards of Accountancy (NASBA)

Python Programming

Online

SOLOLEARN

Certificates: Python for Beginners, Python Intermediate, Python Core, Python Data Structures, Python for Data Science

Machine Learning

Online

DiRAC

2021

DiRAC Virtual AI-athon: This course provided a practical, and hands-on introduction to the concepts, methods, and toolkits for applying machine learning to fundamental scientific problems.

Outreach & Engagement

Teen Tech: Lancashire

JEREMIAH HORROCKS INSTITUTE

UCLan, UK

2023

Developed and facilitated a bespoke activity of exploring the 'Cosmic Fingerprints' of elements in the spectra of celestial objects, as part of the Teen Tech festival.

Virtual Planetarium Shows

JEREMIAH HORROCKS INSTITUTE

UCLan, UK

2020-2021

Planetarium Shows to assist Guides groups with obtaining the Brownies Space badge developed alongside the Royal Astronomical Society. As a joint effort from four JHI academics, we delivered 60 planetarium shows reaching a total of 1 233 girl guides.

Museum of the Moon Exhibition

JEREMIAH HORROCKS INSTITUTE

Harris Museum, UK

2019

The Museum of the Moon is a touring art installation by Luke Jerram. The JHI supported an extended exhibition of the NASA Apollo Missions as well as provided walking tours of the exhibits. I developed the presenter notes and delivered many of the tours.

Engagement Ambassador / Workshop Facilitator

UCLAN/RI YOUNG SCIENTIST CENTRE

UCLan, UK

2016-2024

Facilitation of workshops for high-school classes specializing in 3D Futures, Mission Mars, Life on Mars, Solar System, and Cosmetic Chemistry. Consultation on the development of new physics workshops. Young Scientist Centre Science Demo Competition annual participant and winner in 2016 and 2023.

Alston Observatory

JEREMIAH HORROCKS INSTITUTE

UCLan, UK

2015-2024

Public and community observing evening and visits. Tours of the observatory sites and talks on astronomy. Development of bespoke planetarium talks and activities.

Lancashire Science Festival

JEREMIAH HORROCKS INSTITUTE

UCLan, UK

2015-2024

Exhibitor and developer of the Jeremiah Horrocks Institute stand at the public event which attracts over 5 000 visitors annually.

Cyprus Science Festival

JEREMIAH HORROCKS INSTITUTE

UCLan Cyprus

2016

Development and delivery of an exoplanets activity as well as supporting several other physics activities catering to multiple nationalities and ability levels.

Touching Space

UCLAN PHYSICS SOCIETY

UCLan, UK

2016

Development of astrophysical audio activity and presenter for an event targeted at individuals who have a visual impairment.

Physics Ambassador

OLDHAM SIXTH FORM COLLEGE

OSFC, UK

2014

Representing the college at various public events and open evenings.

Service & Leadership

2016-2024 **UCLan/Ri Young Scientist Centre**, Engagement Ambassador
2020-2022 **Royal Astronomical Society Early Career Network**, Membership Officer
2023 **UK Parliament: Voice of the Future**, Royal Astronomical Society Delegate
2016-2020 **STEM Learning**, STEM Ambassador
2017-2019 **University of Central Lancashire Students' Union**, Chair of Students' Council
2017-2018 **Institute of Physics - Lancashire and Cumbria Branch**, Student Representative
2016-2023 **UCLan Physics Society**, Treasurer (22/23), President (17/18), Secretary (16/17)

Professional Memberships

Fellow of the Royal Astronomical Society, UK (*FRAS*)
Graduate Member of the American Astronomical Society
Member of the Institute of Physics, UK (*MInstP*)
Junior Member of the European Astronomical Society
Ogden Trust Alumni, UK

Junior Associate of LSST:UK (Galaxies; Stars, Milky Way and Local Volume Science Collaboration)
N-body Shop Member